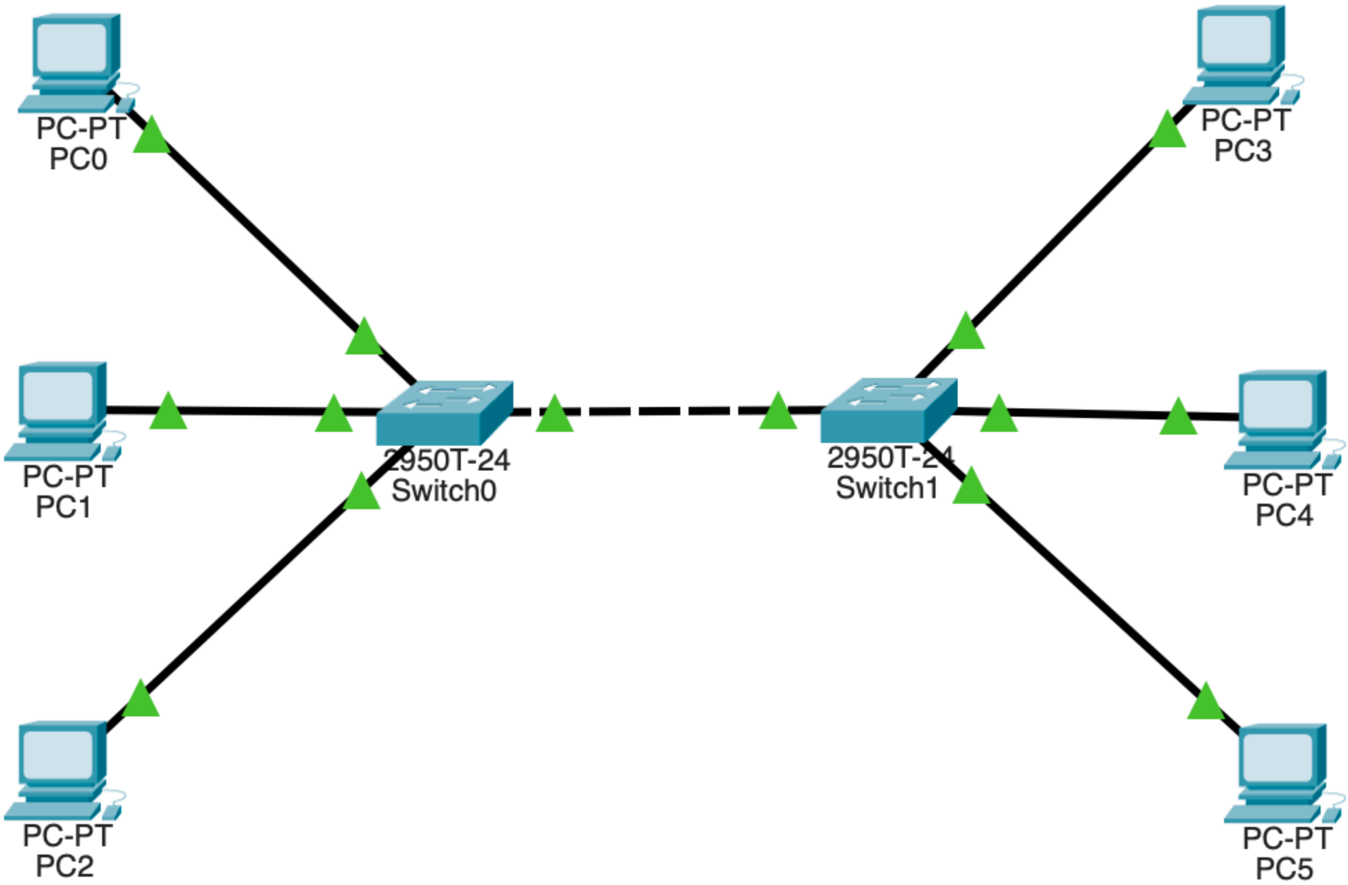
Network Design (Various Cables 2 Router & 6 PC)

1. Group 1 – Left side (Switch0)



Connect:

- `PC0 → Switch0` (use FastEthernet cable)
- `PC1 → Switch0`
- `PC2 → Switch0`

2. Group 2 – Right side (Switch1)

Connect:

- `PC3 → Switch1`
- `PC4 → Switch1`
- `PC5 → Switch1`

3. Connect the two switches

- Use a **crossover cable** (or straight-through, Packet Tracer will auto-correct)
- Connect: `Switch0 (FastEthernet0/24)` → `Switch1 (FastEthernet0/24)`

Assign IP Addresses (Example Setup)

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.1.1	255.255.255.0
PC1	FastEthernet0	192.168.1.2	255.255.255.0
PC2	FastEthernet0	192.168.1.3	255.255.255.0
PC3	FastEthernet0	192.168.1.4	255.255.255.0
PC4	FastEthernet0	192.168.1.5	255.255.255.0

Device	Interface	IP Address	Subnet Mask
PC5	FastEthernet0	192.168.1.6	255.255.255.0

All devices are now in the same **LAN (192.168.1.0/24)** and can communicate through the switches.

Testing Connectivity

1. Click any PC → Desktop tab → **Command Prompt**
2. Type:

   ```
   ping 192.168.1.6
   ```

   (example: from PC0 to PC5)
3. You should see replies — meaning all devices are connected successfully.

Theroy Data

Various Types of Cable

Icon	Cable Name	Use Case (Connects...)	Advantages	Disadvantages
	Automatically Choose Connection Type	Let Packet Tracer decide best cable	Easy for beginners	May choose wrong cable in manual topologies
	Console Cable (RS-232)	PC (Terminal) → Router/Switch (Console Port)	Allows CLI access to configure devices	No data transfer; only used for configuration
	Copper Straight-Through	PC → Switch, Switch → Router	Standard Ethernet cable for different devices	Won't work if used between same devices
	Copper Cross-Over	PC → PC, Switch → Switch	Connects similar devices without hub	Obsolete with auto-MDI/MDI-X
	Fiber Optic Cable	Switch → Switch (long distance), Backbone links	Long distance, high bandwidth, no EMI	Expensive, fragile, harder to install
	Phone Cable	PC → Modem (rare now), DSL lines	Used in telephone modems	Very slow, outdated
	Coaxial Cable	Used in cable modems, older networks	Shielded, less interference	Bulky, lower bandwidth than fiber
	Octal Cable	Router → Multiple terminal ports (Console Access Server)	Manage multiple routers from one PC	Requires special serial interface